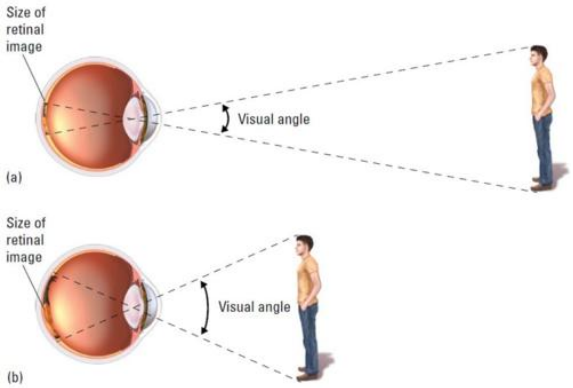
Chapter 62 Optical Instruments
Calculus Physics II

Angular Magnification – is the amount of the human eyes retina the image covers, the more the image covers the eye's retina and the larger the image appears. This can be changed by the aid of lens or mirrors. The actual size of the object does not change but only the image and the size of the image on the retina.

Apparent size – is the size of the image that is produced on the observer's retina (back of observer's eye) by the object and depends on:

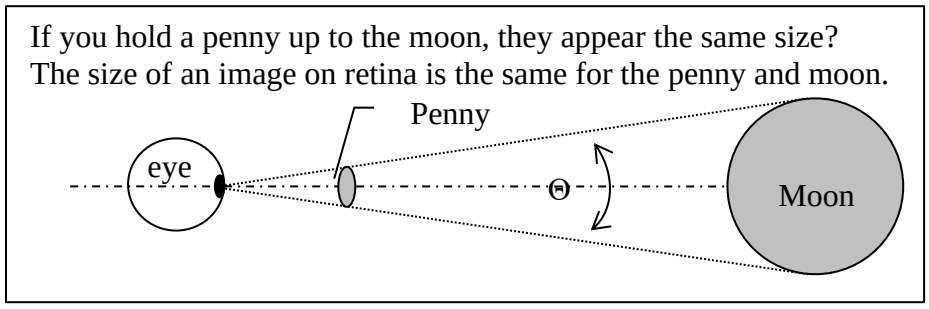
1) The actual size of the object.
2) How far the object is from the observer.
3) Magnification of any optical system being used.



- The apparent size depends on the angle the ray of light from the object makes with the eye.

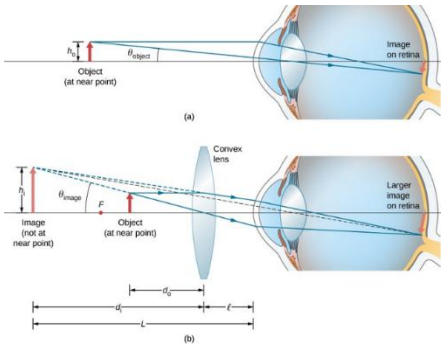
Angular size – The angle an object subtends at the eye of the viewer.

Example of the size of a penny and the size of the moon.

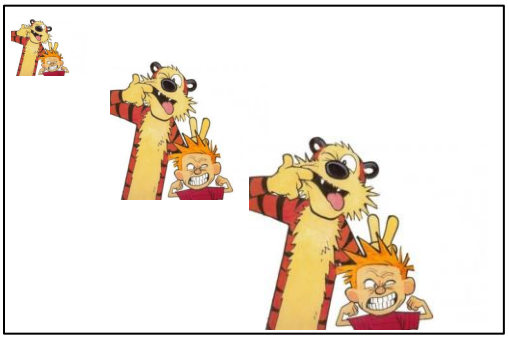


- The image on the retina of the moon and penny are the same size because they both make the same angle entering the eye, illuminating the same area of the retina, but the objects are obviously not the same size!

- The larger the angle that the object makes with the eye the more of the retina is covered and the larger the object appears to be or the closer the object appears to be.
- The smaller the angle that the object makes with the eye the less of the retina is covered and the smaller the object appears to be or is farther away the object appears to be.



Example: for the images of Calvin and Hobbes, are they getting closer to us or is the image size just changing?



- The smaller image covers less of our retina, we interpret the smaller image as farther away.
- The larger image covers more of our retina, we interpret the larger image as closer.

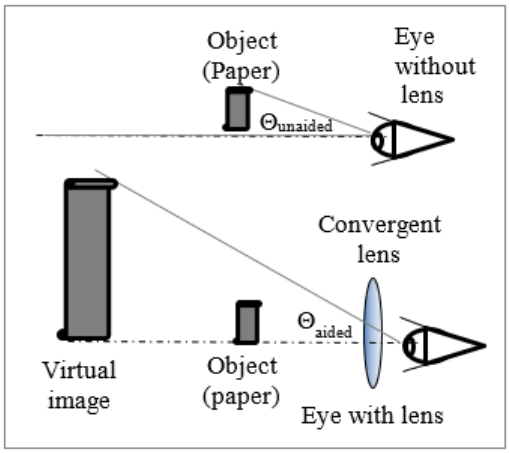
Angular Magnification (M) - The mind interprets the size of an object by the amount of the retina that is covered by the image or the size of the image on the retina.

- Angular Magnification is the ratio of the final angular size produced by an optical instrument to the unaided eye angular size of the object.

$$M = \frac{(\text{Angular size produced by optics}}{\text{reference angular size un} - \text{aided eye}} = \frac{\Theta'}{\Theta}$$

$$M = \frac{\Theta_{\text{aided}}}{\Theta_{\text{unaided}}}$$

Example: If Mike looks at a piece of paper without aid of a lens and the paper makes an angle of $\Theta_{unaided} = 10°$ relative to his eye, then to see the page better he uses a convergent lens and places the page inside the focal length of the lens creating an enlarged virtual image that makes an angle of $\Theta_{aided} = 30°$ relative to his eye, how much was the paper magnified?



$$M = \Theta_{aided}/\Theta_{unaided} = 30°/10° = 3$$

✓ The image is three times bigger due to the lens.

Magnifying Glass - is a convergent lens with a short focal length (high refracting power). The object is placed near but just inside the focal length of the lens, the image formed is virtual, magnified, larger, on the same side of the lens as the object and upright seen by looking into the lens.

- The smaller the focal length the larger the magnification.

Angular magnification of a magnifying glass with the image at infinite.

$$M \approx \frac{N}{f} = \frac{25\text{ cm}}{f}$$

N - is the distance from the eye to the near point (25 cm) for normal vision.

Example: Cathy wants to read a map and uses a magnifying glass (f = 6 cm), if the writing on the map has a font of 4 point (1.4 mm) what size font does she see in the magnifying glass?

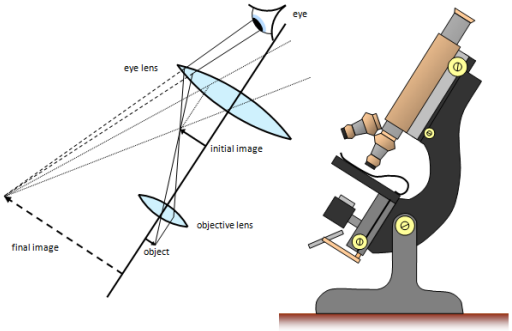


$M = 25\text{ cm}/f = (25\text{ cm})/(6\text{ cm}) = 4.17$

$M = h_i/h_o$ therefore $h_i = Mh_o = (4.17)(1.4\text{ mm}) = 5.8\text{ mm}$

Compound Microscope - An optical device consisting of an objective lens (f_o) and an eyepiece lens (f_e) which produces a larger, inverted and virtual image of an object the is very close to the objective lens.

$$M = -\frac{d_i N}{f_o f_e}$$



f_o - Focal length of objective.
f_e - Focal length of eye-piece.
d_i – image distance formed by the objective lens only.
N - is the distance from the eye to the near point.
N = 25 cm.

- For a microscope the focal length of the objective lens is always shorter than the focal length of the eye-piece. ($f_o < f_e$)
- In a Microscope the object is placed very close to the focal point of the objective lens system.

Objective – lens closest to the object.
Eye-piece – lens that the image is viewed through.

Example: Bob is looking through a microscope that has an objective focal length of 1 cm and an eye piece focal length of 2.5 cm and the object Bob is looking at under the microscope his 1.1 cm from the objective lens, what is the magnification of the microscope?

First the image distance of the object formed by the objective lens needs to be determined.

$f_o = 1\text{ cm}$ and $d_{object} = 1.1\text{ cm}$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \quad solve\ for\ d_i \qquad d_i = \frac{1}{\left[\frac{1}{f_o} - \frac{1}{d_{object}}\right]} = \frac{1}{\left[\frac{1}{1\text{ cm}} - \frac{1}{1.1\text{ cm}}\right]} = +11\text{ cm}$$

Calculating the magnification using the image distance calculated above and known values.

$$M = -\frac{d_i N}{f_o f_e} = -\frac{(11\text{cm})(25\text{cm})}{(1cm)(2.5cm)} = 110$$

Telescope - A device which use a lens/mirror combination to magnify distant objects (objects that are a long way) with the aid of objective and eyepiece lenses, producing a final image which is inverted and virtual.

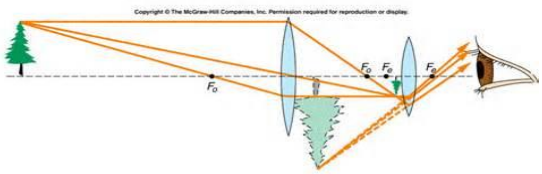
Refracting Telescopes – Telescopes that use a combination of lenses to refract the light producing a larger image from a distant object.

- For a telescope the focal length of the objective lens is always larger than the focal length of the eye-piece. ($f_e < f_o$)
- In a telescope the object is placed very far (near infinity) from the objective lens system.

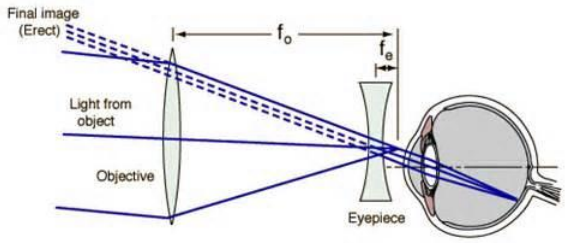
There are two basic types of refracting telescopes.

Astronomical telescope – uses two convergent lenses to produce an enlarged, virtual image that is inverted. This inverted image can be flipped upright by some simple optics.

Astronomical Telescope



Galilean Telescope – uses a convergent lens as the objective and a divergent lens as the eyepiece. The image that is formed is enlarged, virtual and upright.



Magnification of a refracting telescope:

$$M = -\frac{f_{objective}}{f_{eyepiece}}$$

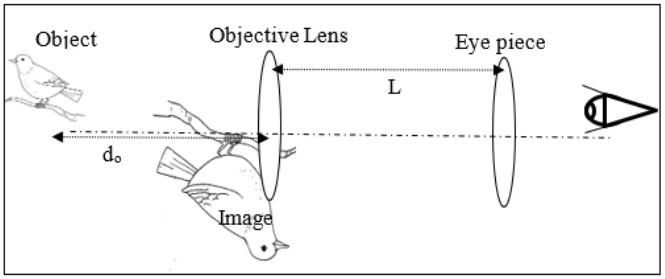
The negative sign tells if the image is inverted or upright

"-" magnification means the image is inverted.
"+" magnification mean the image is upright.

- The advantage to a refracting telescope is that they can be built compact and cost less to manufacture.
- The disadvantages of a refracting telescope are that they cannot be scaled up to larger systems easily and they do not form as clear an image due to chromatic dispersion.

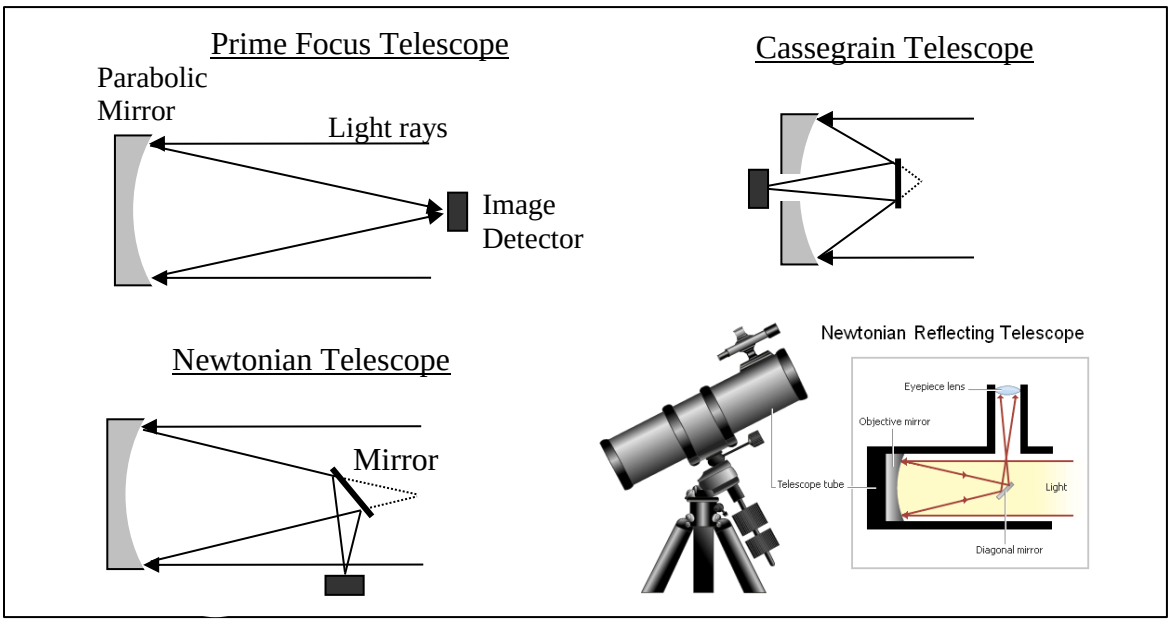
Example: Mike is bird watching with his telescope, his telescope has an objective with a focal length of 120 cm and an eyepiece with a focal length of 3 cm, what is the magnification of Mike's telescope?

$$M = -\frac{f_{objective}}{f_{eyepiece}} = -\frac{120cm}{3\ cm} = 40$$



Reflecting Telescopes – use mirrors to form an image of the objects far away. The larger the reflector, the more light the mirrors collect and the better quality the image.

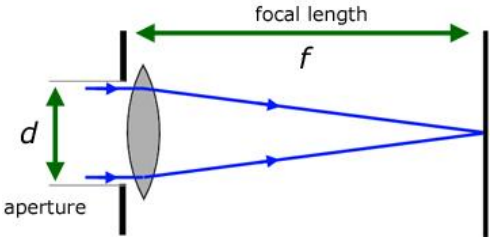
- Advantages of reflecting telescopes are they can be built very large and do not suffer from poor image quality.
- The reflecting telescopes do not suffer from chromatic dispersion, the different wavelengths do not reflect different directions!
- The Hubble and Webb telescopes are Cassegrain style reflecting Telescope.



Camera – is an optical device used to capture images and stores them electronically or on film using a combination of lenses that focus the image on the sensor.

f-number of a lens system (#) - The f-number limits the angle that the rays of light can have as they enter the camera for most cameras an adjustable aperture is placed in the lens system to obtain the different f-numbers.

$$\# = \frac{f}{D}$$



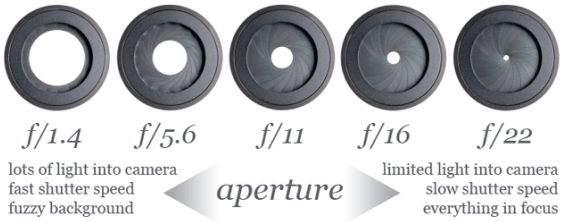
- is the f-number
f - focal length of lens system

D - aperture diameter

f-number is usually formatted f/# where # is the f-number of the lens system.

Example: the camera has f/16, this means that the f-number is 16.

- With a smaller D (large f/#) the light rays are more parallel coming into the camera, so the focal plane is less sensitive to position. In other words, everything is in focus because everything appears to be coming from far away (parallel ray).



- With small f/# (large D) the ray of light can have a larger angle and then the focal plane is much more sensitive to movement, only what is at the focal plane is in focus everything else in front of or behind are out of focus (f/2.8). This is shown in the top left image below. For large f/# (small D) the light is almost parallel coming into the camera therefore all the light is imaged at the image plane, everything is in focus. This is shown in the lower left image below (f/16).

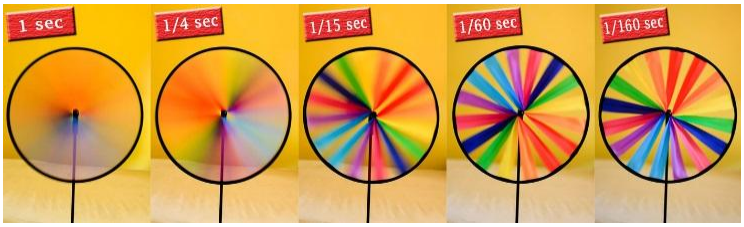


Shutter speed – is the amount of time the film or sensor is exposed to the light from the image of the camera.

$$Shutter\ speed = \frac{1}{Exposure\ time}$$

Example: a shutter speed of 1/15 s the shutter is open for $\frac{1}{15s}$ = 0.067 /seconds

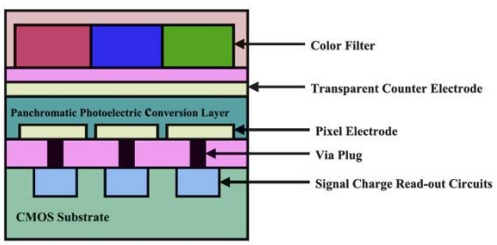
- ❖ The faster the shutter speed the less motion appears of the photographic image, this creates a sharp image.
- ❖ The slower the shutter speed the more motion appears on the photographic image, this creates a burred image.
- ❖ The faster shutter speeds the less light exposure on the sensor and the slower shutter speeds the more total light exposure for the sensors.

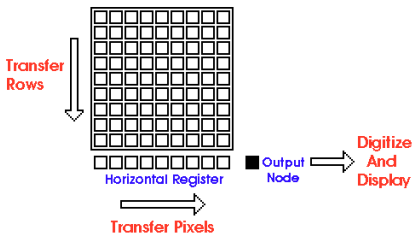


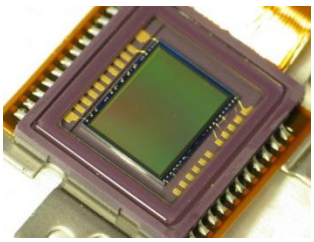
Digital Cameras - CMOS (Complementary Metal-Oxide-Semiconductor) for imaging- Digital cameras primarily use CMOS sensors to capture images, although some high-end or specialized cameras use CCD (Charge-Coupled Device) sensors. Both types are solid-state semiconductor sensors that convert light (photons) into electrical signals (electrons) using an array of pixels (photodiodes).

CMOS Sensors: Found in most modern smartphones, DSLR, and mirrorless cameras. They are preferred because they consume less power, are cheaper to manufacture, and process data faster than CCD sensors.

The sensors use an array of photo diodes to convert the incident light into electric charge and store that charge on tiny capacitors, the more light that is incident on the photodiode the more charge is transferred to the capacitors. The charge is then detected and converted to an electronic image.





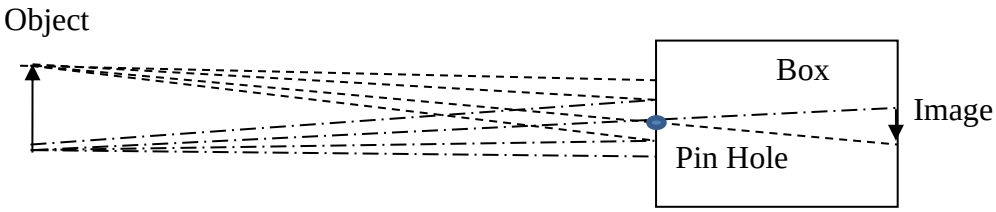


Auto focus (AF) in cameras Active and passive

Active AF – use a beam of sound or infrared light to gauge distance then sets the focus at that distance.

Passive AF – uses the sensors array to determine the contrast of the image between pixels. If the image is burry then there is not much contrast the adjacent pixels will not change much in intensity but with high contrast the pixels will have larger variation and high contrast means the image is in focus.

Pin Hole "Lens" or Camera - A real inverted image can be formed with a very small pin hole and a box. The small hole blocks all rays that are not paraxial (nearly parallel) therefore only light coming directly from a single point on the object gets to the back of the box (image plane).

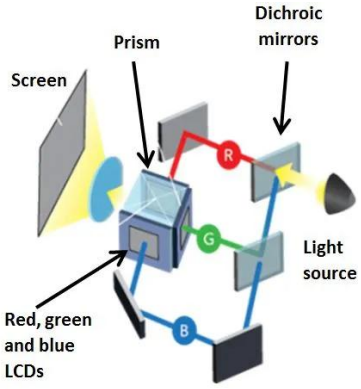


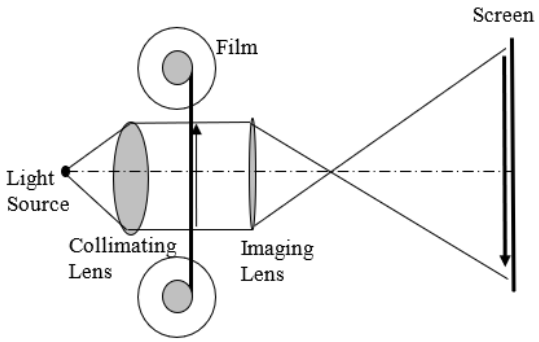
Examples of pictures taken with a pinhole camera.



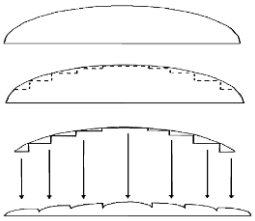


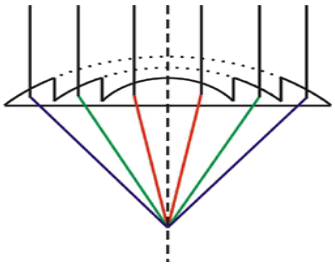
Projector – A projector uses a light source that is collimated by a condensing lens, this collimated light goes through the object (film or digital camera is the object) and a real image of this object is magnified, and the image plane is formed at the screen.



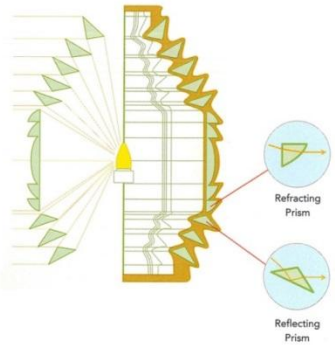


Fresnel Lens – A refracting convergent lens is made into a Fresnel Lens by "flattening" the lens into conical sections, these lenses are used as condensers to collimate point sources of light into parallel beams of light. Because they are flat, very large lens can be made that are lighter weight and less costly.





Examples: light house lens, overhead projection lens.



Cutaway illustration of Fresnel's lighthouse-lens system.
Some of the prisms in a Fresnel lens refract the light, allowing it to pass through in the desired direction. Some of the prisms reflect the light internally off one face, completely changing its course before it exists.





- Fresnel lenses are generally not good lenses for imaging.

Fiber Optics – Optical fibers are long strands of glass or plastic fiber that transmit light using total internal reflection of the light. The light is total internally reflected down the fiber at the core/cladding interface so there is very little loss of light as it moves down the length of the fiber. Light can travel very long distances using optical fiber.

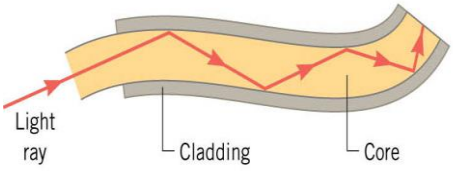
The minimum angle to achieve total internal reflection is called the critical angle (Θ_c)



$$\Theta_C = \sin^{-1}\left(\frac{n_{clad}}{n_{core}}\right)$$

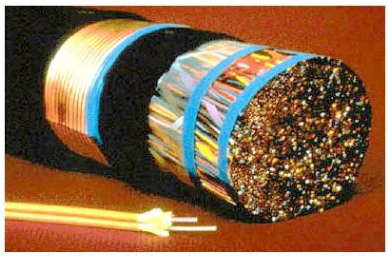
Core – Center portion of the fiber that is made of optically transparent material with a high index of refraction.

Cladding – optically transparent material that surrounds the core that has a lower index of refraction than the core.



- ❖ Light entering the fibers core experiences total internal reflection at some angle larger than the critical angle (measured from the normal to the end surface of the fiber) for the core/clad interface. The light is total internally reflected down the fiber length.

Because fiber optics uses light and not electrons to transmit information it can transmit much more information in a lot shorter time. Fiber optic cables have replacing copper cables because of their high band width (capable of transmitting a lot of information) capability. The figure shows the small fiber optic cable in the fore ground and the large copper cable, the fiber optic cable has the same if not more capacity to transmit information than the whole copper cable.

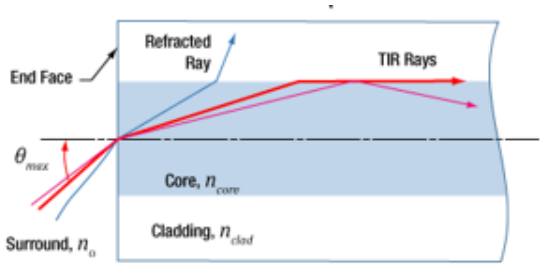


The optical fiber cable in the foreground has the equivalent capacity of the copper cable in the background.

Numerical Aperture (NA) - a dimensionless number in fiber optics that quantifies the light-gathering ability of an optical fiber. It describes the maximum angle at which light can enter the fiber's core and still be guided by the principle of total internal reflection.

$$NA = n_{out}\sin(\theta_{max}) = \sqrt{(n_{core})^2 - (n_{clad})^2}$$

NA – Numerical aperture
Θ_{max} – maximum acceptance angle
n_{out} – index of refraction outside the fiber surface.
n_{core} – index of refraction of the core material.
n_{clad} - index of refraction of the clad material.



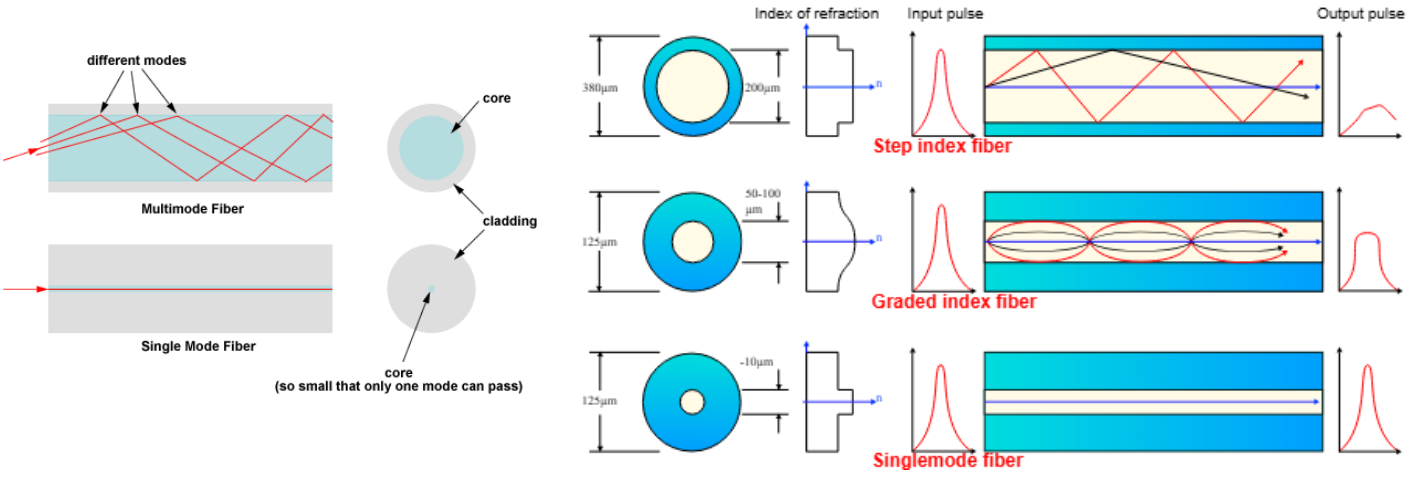
Single-mode and multimode are two types of fiber optic cables, distinguished primarily by the size of their core and the number of light paths, or "modes," they can carry. Single-mode fiber is ideal for long-distance, high-bandwidth applications, while multimode is a more cost-effective solution for shorter-distance needs.

Single-mode fibers are designed with a small core and low NA to guide only a single mode of light. This eliminates intermodal dispersion but requires very precise light coupling.

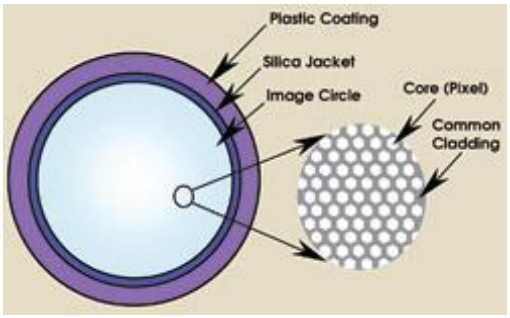
Multimode fibers have a higher NA and larger core, allowing for simpler coupling but a lower bandwidth over long distances.

Step index fiber - is a type of optical fiber with a uniform refractive index in the core and an abrupt decrease in refractive index at the core-cladding boundary. This "step" in refractive index causes light to travel by undergoing total internal reflection, but in multimode fibers, different light paths can cause modal dispersion, where light pulses spread out, limiting bandwidth and distance.

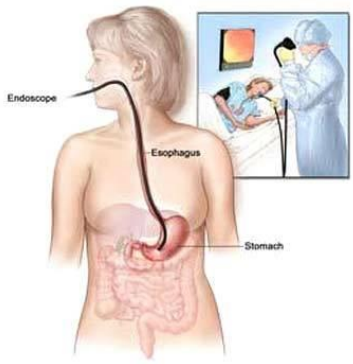
Graded index fiber - A graded-index (GRIN) fiber is a type of optical fiber in which the refractive index of the core gradually decreases from the center toward the outer edge. This unique profile is designed to correct for modal dispersion, a common signal distortion problem in multimode fibers, allowing for higher bandwidth and transmission speeds.



Fiber Bundles for imaging – A single fiber can not transmitt an image but if many small fibers (20 μm diamter or less) are bundled together, each fiber transmittes a spot (pixel) of light that is tramitted to the other end of the fiber, the combination of all the fiber transmitt a complete image from one end of the fiber bunble to the other.



Fiber optic endoscopes use fiber bundles to image inside of the body.





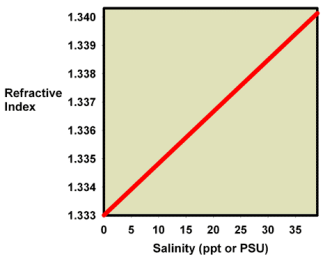
Refractometer – A refractometer uses total internal reflection of light to measure the index of refration of a substance in contact with the glass surface of the instrument. The substance has a slightly different index of refraction than the glass, this different index of refraction changes the critical angle for total internal reflection. The Refratometer measures this shift in angle and the change in index of refraction is calibarated to the consintration of sugar or salt in water. Therefore the refractometer can measure concentrations of surgar or salt in a soluiton by mearsuring its index of refration, the more sugar or salt in water the higher the index of refratction.

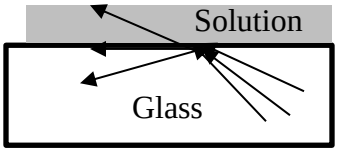


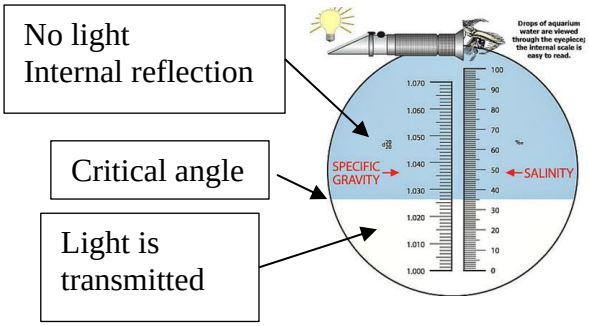
Degrees Brix – is the measure of the amount of surgar in a solution, 1 degree Brix is 1 gram of sugar in 100 grams of solution.

Salinity – is the measure of the amount of salt in a solution and is given in parts per thousand (PPT) or parts per million (PPM)

Example: A refractometer is used to measure the concentration of a solution such as the amount of salt in water. If $n_{glass} = 1.55$ and the refractometer measures the critical angle between the salt solution and glass plate to be $\Theta_c = 59.61°$, what is the index of refraction of the solution

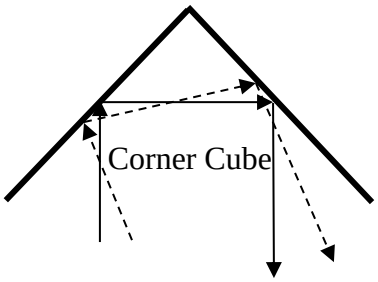


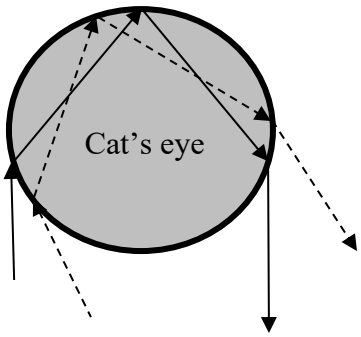




$n_{solution} = n_{glass}\sin(\Theta_c) = 1.337$ and from the graph this corropsones to 25 ppt

Retro reflectors – light rays are reflected back the same direction they came from; any light incident of the retro reflector will be reflected back to the source of the light.





Applications for retro reflectors include reflectors for bikes and cars, safety reflectors on clothing and cats eyes on lines on the road so they "light up" from your cars head lights.

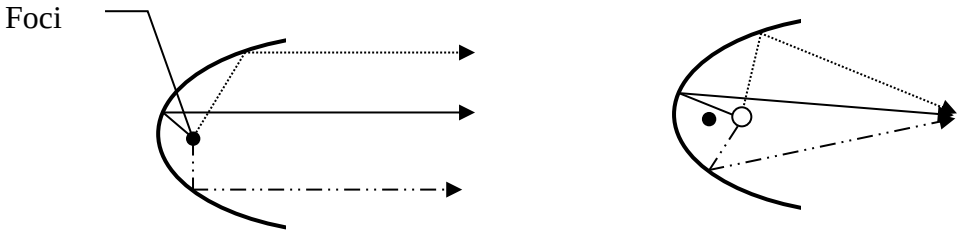






Parabolic Reflector – Light placed at the foci will be reflected out parallel. If the light is placed just past the foci then the light will be focused at a point in front of the mirror

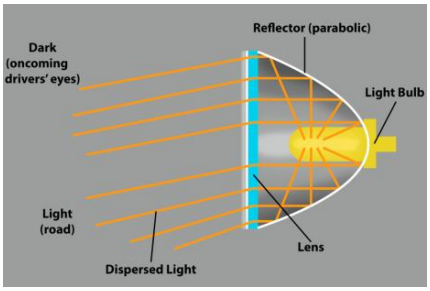
Parabolic Reflector



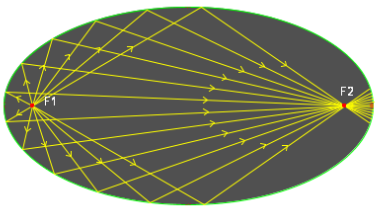
Applications for Parabolic Reflectors are focusing EM waves for antennas and collimate light for head lights on cars

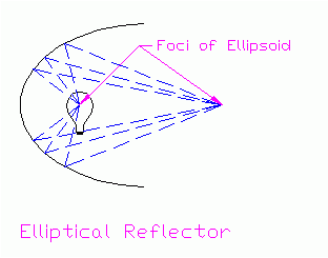






Elliptical Reflector – If a light source placed at one of the foci all the light will be focused at the other foci.





Applications for elliptical refectors are focuing light at a point (dental light) and spot lights in theaters

